

PAUL ALEX SAMUEL JOHN

AI Engineer · LangGraph · Azure OpenAI · Multi-Agent Systems · RAG Pipelines · 2 YOE

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years of production experience building enterprise Generative AI systems, multi-agent LLM platforms, and RAG pipelines. Deployed end-to-end AI products at JSW Steel (Top-500 India enterprise) — including a 6-agent Text-to-SQL platform processing 500+ daily queries and a hybrid RAG system improving retrieval precision by 35%. Sole AI Engineer on a cross-functional team, owning full product lifecycle from architecture through CI/CD deployment. Core stack: Azure OpenAI · GPT-4 · LangGraph · Semantic Kernel · Pinecone · Azure AI Search · FastAPI · Docker · Kubernetes (AKS).

EXPERIENCE

AI Engineer | JSW Steel (via Sereno)

May 2024 – Present · Mumbai, India

Sole AI Engineer embedded with JSW Steel's enterprise team, independently architecting and delivering production GenAI systems across business units.

- Architected a 6-agent autonomous Text-to-SQL platform (LangGraph + Azure OpenAI GPT-4 + Azure AI Search) handling 500+ daily enterprise queries — cutting analyst SQL reporting time by 60% with 87% first-pass query accuracy.
- Built a production hybrid RAG system (dense + BM25 + cross-encoder reranking) with parent-child chunking and HyDE query expansion, boosting retrieval precision by 35% over naive baseline; deployed on Azure AI Search and Pinecone.
- Deployed containerized AI inference microservices on Azure Kubernetes Service (AKS) using Docker + async FastAPI with connection pooling — sustaining sub-300ms P95 latency under concurrent enterprise load.
- Developed a Kannada-to-Hindi speech translation pipeline (Whisper ASR + Helsinki-NLP NMT + Coqui TTS) achieving sub-3s end-to-end latency and BLEU score of 28+ on held-out test corpus.
- Owned full AI product lifecycle across 4+ business units: requirements gathering, architecture design, prompt engineering, vector indexing, retrieval tuning, CI/CD deployment, and production monitoring.

AI Engineer Intern | NxtGen Cloud Services

Jan 2024 – May 2024 · Bengaluru, India

- Constructed async AI inference APIs (FastAPI + Azure + SageMaker) with request batching and connection pool optimization — cutting average inference latency by 40% (850ms → 510ms) across multi-model enterprise workloads.
- Built an enterprise document AI ingestion pipeline processing 1,000+ PDFs/day via Azure Document Intelligence + OpenAI text-embedding-3-small + Azure Cognitive Search with hybrid retrieval and metadata filtering.
- Established CI/CD pipelines (GitHub Actions + Azure DevOps) for blue-green AKS deployments and automated model versioning — reducing deployment cycle time from 2+ hours to under 15 minutes.
- Integrated Pinecone and ChromaDB vector databases with embedding pipelines and reranking layers, improving enterprise knowledge retrieval accuracy by 30% over keyword-only baselines.

PROJECTS

Multi-Agent NLP-to-SQL Enterprise Platform

Azure OpenAI GPT-4 · LangGraph · Azure SQL · Cosmos DB · FastAPI · Docker · AKS

- Architected a production 6-agent Text-to-SQL system for JSW Steel (schema retrieval → semantic routing → SQL generation → validation → refinement → answer synthesis) achieving 87% first-pass accuracy on enterprise analytics queries.
- Implemented multi-turn session memory via Cosmos DB; deployed on AKS serving 500+ daily queries at sub-300ms latency with async FastAPI and streaming response support.

Enterprise Document Intelligence & Semantic Search Pipeline

Azure Document Intelligence · OpenAI text-embedding-3-small · Azure Cognitive Search · FastAPI

- Created a production OCR-to-semantic-search pipeline ingesting 1,000+ enterprise PDFs/day; implemented parent-child chunking, HyDE expansion, and cross-encoder reranking — achieving 35% retrieval precision improvement and sub-400ms API response latency.

Semantic Kernel Autonomous Agent System

Semantic Kernel · GPT-4 · Pinecone · Hugging Face · FastAPI

- Developed a modular multi-step agent system with plugin-based tool architecture, Pinecone vector memory for long-horizon context, and dynamically configurable LLM backends (GPT-4 / local Hugging Face models) for cost-optimized deployment.

Kannada-to-Hindi Real-Time Multilingual Speech Translation

Whisper ASR · Helsinki-NLP · Coqui TTS · Hugging Face · FastAPI

- Sole engineer for a full speech-to-speech pipeline achieving sub-3s latency and 28+ BLEU score; responsible for architecture design, evaluation methodology, and cloud deployment.

TECHNICAL SKILLS

Generative AI & LLM: GPT-4, GPT-4o, Azure OpenAI, LangChain, LangGraph, Semantic Kernel, RAG, Agentic AI, Multi-Agent Systems, Prompt Engineering, Function Calling, Structured Outputs, MCP

AI / ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, BERT, Whisper, Fine-tuning, LoRA, PEFT, ONNX

Vector & Search: Pinecone, ChromaDB, FAISS, Azure AI Search, Weaviate, Hybrid Retrieval, Cross-Encoder Reranking, HyDE

Cloud & DevOps: Azure AI, Azure ML, AWS SageMaker, GCP Vertex AI, Docker, Kubernetes (AKS), CI/CD, GitHub Actions, Azure DevOps

Backend & APIs: Python, FastAPI, Flask, REST APIs, Async Python, SQL, Cosmos DB, PostgreSQL, Redis

Document AI & NLP: Azure Document Intelligence, OCR Pipelines, NER, Information Extraction, OpenCV, ASR, TTS

EDUCATION

B.Tech in Computer Science Engineering (AI & ML)

2020 – 2024

Avanthi St. Theresa College of Engineering & Technology, Hyderabad

CERTIFICATIONS

- AWS Certified Machine Learning Specialty
- Microsoft Azure AI Fundamentals (AI-900)
- Google Cloud Professional Machine Learning Engineer
- Hugging Face Agent Tools Certification